

Luzian Merlin Rhys Seeburger

Github: github.com/RhysSeeburger

Website: rhysseeburger.github.io/

Nationality: Austrian

Liverpool John Moores University, Liverpool

Email: seeburger.astro@gmail.com

LinkedIn: linkedin.com/in/rhys-seeburger/

DOB: 25.09.1997

SUMMARY

Post Doctoral Research Assistant at the Astrophysics Research Institute at LJMU, researching dormant black holes in close binaries. Also interested in progenitor systems and confounders (e.g. post-interaction systems). Keenly engaged in science education and communication and always willing to share my passion for astrophysics with others.

EDUCATION

- Max Planck Institute for Astronomy & Heidelberg University** Heidelberg, Germany
IMPRS Fellow, PhD - Astronomy; Magna cum Laude September 2021 - July 2025
- University of Edinburgh** Edinburgh, United Kingdom
Master of Physics - Astrophysics; First Class September 2016 - June 2021
- Sacre Coeur Wien** Vienna, Austria
Matura; Average: 1.0 September 2008 - June 2016

SELECTED HONORS AND AWARDS

- Emerson Prize (highest grades out of all graduating Astrophysics students) - 2020/21
- Integrated Masters Medal for Astrophysics - 2020/21
- Senior Honours Medal for Astrophysics - 2019/20

SKILLS SUMMARY

- Languages** Python (Advanced), Bash, C++, Julia, Fortran (Basic)
- Tools & IDEs** Git, MESA, Anaconda, Docker, WSL, VSCode, Atom, Spyder
- IT - general** Machine Learning (Scikit, PyTorch), High Performance Computing, Microsoft Office, Keynote, Prezi
- Soft Skills** Leadership, Scientific Writing, Public Speaking

TALKS

- Match made in Heaven: Gaia and APOGEE for binary analysis** Birmingham, UK
National Astronomy Meeting 07/26
- The curious case of NGC 1850 BH1** Liege, Belgium
To Be or not to Be: Understanding the star, the disk, and the stripped companions 07/26
- Match made in Heaven: Gaia and APOGEE for binary analysis** Leuven, Belgium
SDSS-V Collaboration Meeting 07/26
- The curious case of NGC 1850 BH1** Liverpool, UK
Friday Afternoon Research Talk, Liverpool John Moores University 06/26
- It takes two to tango: Binary Spectroscopy of NGC 1850** Potsdam, Germany
MUSE Busyweek 04/26
- Thoughts on (stable) mass transfer & outcomes** Santa Barbara, USA
Stellar-Mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys 11/25
- A detailed analysis of a sample of black hole impostoring post-mass-transfer binaries** Santa Barbara, USA
The Lifecycle of Stellar Black Holes 11/25
- Untangling binary star systems with SDSS-V APOGEE spectra** Lijiang, China
Binary Stars in a New Era 08/25
- A detailed analysis of a sample of black hole impostoring post-mass-transfer binaries** Edinburgh, UK
Coffee talk, Royal Observatory Edinburgh 10/24
- A detailed analysis of a sample of black hole impostoring post-mass-transfer binaries** Glasgow, UK
Glasgow Astronomy Seminar, Glasgow University 10/24

- **A detailed analysis of a sample of black hole impostoring post-mass-transfer binaries** Leuven, Belgium
IvS Seminar, KU Leuven 10/24
- **Black Hole Imposters** Leiden, Netherlands
From discovery to a population: benchmarking stripped stars and companions 07/24
- **A detailed analysis of a sample of black hole impostoring post-mass-transfer binaries** Liege, Belgium
The Eventful Life of Massive Star Multiples 07/24
- **Autonomous Disentangling** Online
GALAH Science Telecon 06/24
- **Black Hole Companions** Ringberg, Germany
Galaxies & Cosmology Department Retreat 05/24
- **Spectral Disentangling for Large Surveys** Online
Kansas University Seminar 02/24
- **Hunting for Dark Companions** Garching, Germany
Stellar Astrophysics in the Era of Gaia, Spectroscopic, and Asteroseismic Surveys 08/23
- **Taking Spectral Disentangling to the Industrial Scale** Krakow, Poland
EAS Annual Meeting 07/23
- **Hunting for Dark Companions** Mitzpe Ramon, Israel
The Time Domain in the Era of Astronomical Big Data, Wise Observatory 02/23
- **Finding Black Holes that are actually Black** Heidelberg, Germany
Galaxy Coffee, MPIA 11/22
- **Any Star could be a Binary** Garching, Germany
The Impact of Binaries on Stellar Evolution, MPA 05/22
- **Disentangling Composite Spectra** Heidelberg, Germany
IMPRS Seminar, University of Heidelberg 05/22

TEACHING EXPERIENCE

- **Tutoring - Advanced Physics Lab for Physicists** Heidelberg University
F30 - Stellar CCD photometry May 2021 - November 2022
- **Ogden Trust Teach Physics Internship** Impington Village College
Including lesson planning, tutoring, small-group teaching June 2021 - July 2021
- **Science Education Placement** Stewart's Melville College
"The effect of Specific Gestures on Learning Physics" September 2019 - December 2019

ACCEPTED PROPOSALS

As Principal Investigator:

- **UVES observations of (post-) mass transfer binaries.** 15 hours
Multi-epoch spectroscopy of 7 Gaia SB1 systems proposed to be (post-) mass transfer SB2 systems. Cycle P115
- **FEROS observations of (post-) mass transfer binaries.** 15 hours
Multi-epoch spectroscopy of 2 Gaia SB1 systems proposed to be (post-) mass transfer SB2 systems. Cycle P115
- **FEROS observations of lower-mass (post-) mass transfer binaries.** 60 hours
Multi-epoch spectroscopy of 8 Gaia SB1 systems proposed to be (post-) mass transfer SB2 systems. Cycle P114
- **FEROS observations of (post-) mass transfer binaries.** 60 hours
Multi-epoch spectroscopy of 6 Gaia SB1 systems proposed to be (post-) mass transfer SB2 systems. Cycle P112

As Co-Investigator:

- **Hunting for stellar-mass black holes in low-metallicity binaries.** PI: El-Badry 150 hours
Observations of ~ 500 low-metallicity stars in astrometric binaries to determine RVs Cycle P116
- **Novel Gaia Candidates for dormant BH or NS binaries.** PI: Müller-Horn 200 Hours
Multi-epoch spectroscopy of 261 giant stars to search for dormant compact companions Cycle P115

- **Hunting for stellar-mass black holes in low-metallicity binaries.** PI: El-Badry 180 hours
Observations of 379 low-metallicity stars in astrometric binaries to determine RVs Cycle P115
- **Novel Gaia Candidates for dormant BH or NS binaries.** PI: Müller-Horn 175 Hours
Multi-epoch spectroscopy of 72 giant stars to search for dormant compact companions Cycle P114
- **Hunting for stellar-mass black holes in low-metallicity binaries.** PI: El-Badry 180 hours
Observations of 379 low-metallicity stars in astrometric binaries to determine RVs Cycle P114
- **FEROS observations of compact objects in astrometric binaries.** PI: El-Badry 180 Hours
Continuation of RV monitoring campaign of astrometric binaries containing compact companions Cycle P112

OUTREACH

- **Activities**
 - STEM Professional, Royal Society Project “Can we get Humans to Mars?” 2026
 - Volunteer at Explore Science Mannheim 2023, 2024, 2025
 - MPIA Outreach Fellow 2022-2025
 - “Tag der Naturwissenschaften” (Science Day) at Experimenta Heilbronn - Guest Scientist 2022
- **Publications**
 - Seeburger, L. (2023) Mein Nachbar, das Schwarze Loch. Sterne und Weltraum, 02/23
 - Seeburger, L. (2022) Wolkig mit Aussicht auf Edelsteinregen. Sterne und Weltraum, 08/22
 - Seeburger, L. (2022) 5000 Augen erfassen die größte Galaxienkarte der Welt. Sterne und Weltraum, 07/22
- **Talks**
 - ChatGPT & Co in der Astronomie Astro & Co, Online, 05/25
 - Schwarzen Löchern auf der Spur Astronomischer Arbeitskreis, Pollichia Museum, Bad Dürkheim, Germany, 01/25
 - Schwarzen Löchern auf der Spur Astronomie am Nachmittag, Planetarium Mannheim, Mannheim, Germany, 10/24
 - Mein Nachbar, das Schwarze Loch Faszination Astronomie, Haus der Astronomie, Heidelberg, Germany, 05/24
 - Das Nächstgelegene Schwarze Loch MPIA Open Day, Heidelberg, Germany, 10/23
 - Mein Nachbar, das Schwarze Loch Faszination Astronomie Online, 03/23
 - Searching for life on other planets 100 Hours of Astronomy, Online, 09/21

SERVICE

- **Academic**
 - Co-Organiser, BinStars Session at NAM in Birmingham 2026
 - LOC SDSS-V collaboration meeting 2025
 - Distributed Peer Review, ESO P115 2024
- **Mentoring**
 - Co-Supervised high school Student’s BOGY internship 2023
 - Peer Support Leader, Physics Peer Mentoring Scheme 2019/20
- **Representation and Leadership**
 - Member of MPIA Equality Committee 2024-2025
 - IMPRS generation Representative 2021-2025
 - School Convener for School of Physics and Astronomy 2020/21
 - Chair for Staff Student Liaison Committee 2020/21
 - Student Representative for Degree Year Group 2020/21
- **Equity, Diversity & Inclusion**

- **Co-Lead Queer International Meetup Heidelberg 2023-2025**
- **Organised “OUTer Space”, LGBT+ Outreach Day at Haus der Astronomie Heidelberg 2023**
- **Gave multiple talks on EDI Topics**
 - * **You can’t be what you can’t see** EAS Annual Meeting, Krakow, Poland, 07/23
 - * **International Day against Homophobia, Biphobia and Transphobia** MPIA, Heidelberg, Germany, 05/23
 - * **LGBTQIA+ in Astronomy** GC Retreat, Schöntal, Germany, 05/23
 - * **LGBT+ Astronomy Icons** OUTer Space, Haus der Astronomie, Heidelberg, Germany, 02/23
 - * **Equity, Diversity and Inclusion in Astronomy** IMPRS Retreat, Salzburg, Austria, 09/22
- **Organised LGBT+ Lunch meetings at School of Physics and Astronomy 2020/21**
- **Member of Edinburgh University Sports Union Inclusion Committee 2020/21**
- **Delegate for UofE for NUS LGBT+ Conference in Edinburgh 2018**
- **Participated in LGBTYS Youth Summit (Falkirk) and NTC (Durham) 2016/17**